DASHAN AND PYTHAGOREAN TRIPLETS

HackerRank

Dashan has been assigned to find the number of Pythagorean triplets from zero to a given number n such that the difference between the two biggest numbers is one i.e., if a, b and c are three numbers such that: $c^2 = a^2 + b^2$ and $b > a$, $c \le n$ then $c - b = 1$.
But Dashan has to prepare for his mid semester examination and he needs your help.
Note: Smallest pythagorean triplet is (3, 4, 5).

Input Format

The first line contains a single integer T - the number of test cases
The next T lines contains a single integer n – as described in the problem

Constraints

T <= 1000000
5 <= n <= 1000000

Output Format

Print a single integer for each test case, the number of required triplets such that the value of c is less than or equal to n.

Sample Input 0

```
1
15
```

Sample Output 0

```
2
```

Explanation 0

There are 2 triplets (3,4,5), (5,12,13) which satisfy given conditions. In first triplet c = 5(<=15) and b = 4(=c-1). In second triplet c= 13(<=15) and b = 12(=c-1).

Sample Input 1

```
2
20
30
```

Sample Output 1

```
2
3
```

Explanation 1

First testcase, (3,4,5), (5,12,13) satisfy given conditions. For second testcase, (3,4,5), (5, 12, 13) and (7, 24, 25).